

Mixed / Multilevel Models

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<https://frodriguezsanchez.net>

Example dataset: trees

- Data on 1000 trees from 10 sites.

```
head(trees)
```

	site	dbh	height	sex	dead
1	4	29.68	36.1	male	0
2	5	33.29	42.3	male	0
3	2	28.03	41.9	female	0
4	5	39.86	46.5	female	0
5	1	47.94	43.9	female	0
6	1	10.82	26.2	male	0

Example dataset: trees

- Data on 1000 trees from 10 sites.
- Trees per site: 4 - 392.

```
head(trees)
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1	4	29.68	36.1	male	0
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Q: What's the relationship
between tree diameter and
height?

A simple linear model

```
lm.simple <- lm(height ~ dbh, data = trees)
```

Call:

```
lm(formula = height ~ dbh, data = trees)
```

Residuals:

Min	1Q	Median	3Q	Max
-13.3270	-2.8978	0.1057	2.7924	12.9511

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	19.33920	0.31064	62.26	<2e-16 ***
dbh	0.61570	0.01013	60.79	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

$$y_i \sim N(\mu_i, \sigma^2)$$

$$\mu_i = \alpha + \beta x_i$$

In this case:

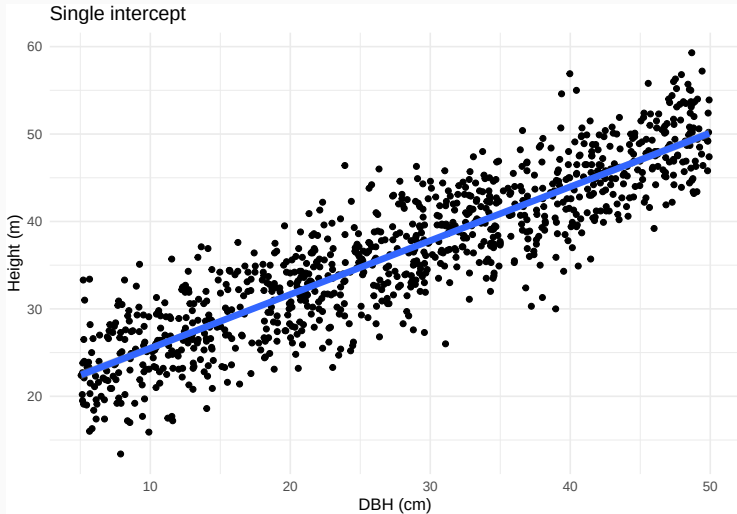
$$\text{Height}_i \sim N(\mu_i, \sigma^2)$$

$$\mu_i = \alpha + \beta \text{DBH}_i$$

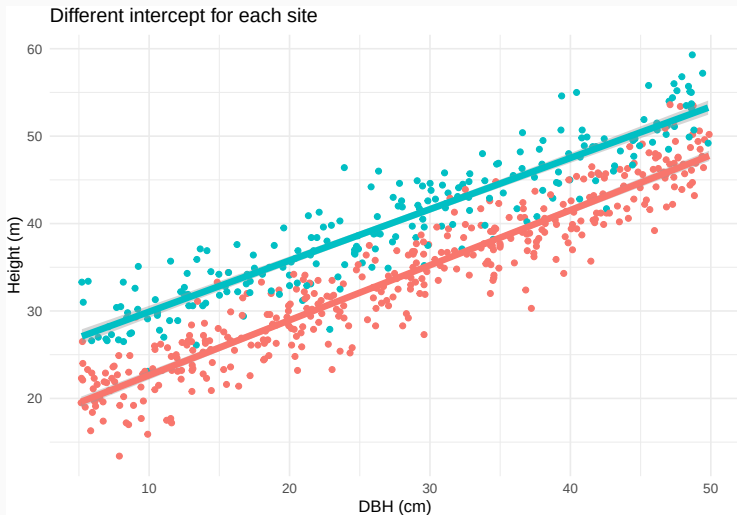
α : expected height when DBH = 0

β : how much height increases with every unit increase of DBH

There is only one intercept



What if allometry varies among sites?



Fitting a varying intercepts model with `lm`

Call:

```
lm(formula = height ~ site + dbh, data = trees)
```

Residuals:

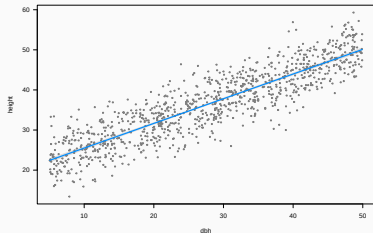
Min	1Q	Median	3Q	Max
-10.1130	-1.9885	0.0582	2.0314	11.3320

Coefficients:

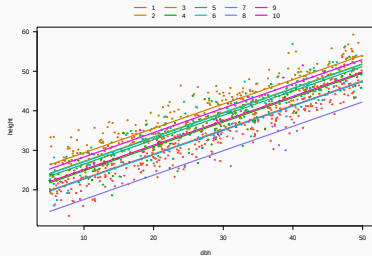
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	16.699037	0.260565	64.088	< 2e-16 ***
site2	6.504303	0.256730	25.335	< 2e-16 ***
site3	4.357457	0.354181	12.303	< 2e-16 ***
site4	1.934650	0.356102	5.433	6.98e-08 ***
site5	3.637432	0.339688	10.708	< 2e-16 ***
site6	4.204511	0.421906	9.966	< 2e-16 ***
site7	-0.176193	0.666772	-0.264	0.7916
site8	-5.312648	0.893603	-5.945	3.82e-09 ***
site9	5.437049	1.087766	4.998	6.84e-07 ***
site10	2.263338	1.369986	1.652	0.0988 .
dbh	0.617075	0.007574	81.473	< 2e-16 ***

Single vs varying intercept

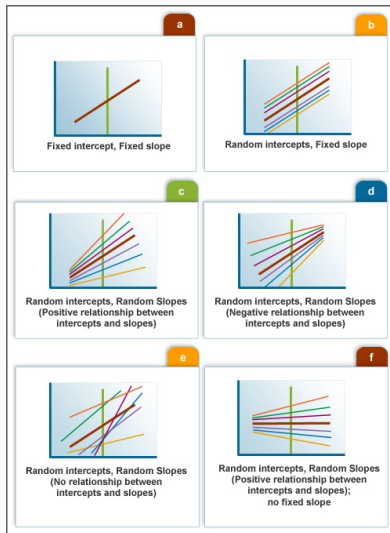
Single intercept



Different intercept for each site



Mixed models enable us to account for variability



$$y_i = a + \alpha_j + b \cdot x_i + \varepsilon_i$$

$$\alpha_j \sim N(0, \tau^2)$$

$$\varepsilon_i \sim N(0, \sigma^2)$$

In our example:

$$\text{Height}_i = a + \text{site}_j + b \cdot \text{DBH}_i + \varepsilon_i$$

$$\text{site}_j \sim N(0, \tau^2)$$

$$\varepsilon_i \sim N(0, \sigma^2)$$

Mixed models estimate **varying parameters**

(intercepts and/or slopes)

with pooling among levels

(rather than considering them fully
independent)

Hence there's gradient between

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 - `lm (height ~ dbh + site)`

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 - `lm (height ~ dbh + site)`
- **partial pooling:** Inter-related intercepts.

Hence there's gradient between

- **complete pooling:** Single overall intercept.
 - `lm (height ~ dbh)`
- **no pooling:** One *independent* intercept for each site.
 - `lm (height ~ dbh + site)`
- **partial pooling:** Inter-related intercepts.
 - `lmer(height ~ dbh + (1 | site))`

1. Fixed effects **constant** across individuals, random effects **vary**.

http://andrewgelman.com/2005/01/25/why_i_dont_use/

Random vs Fixed effects?

1. Fixed effects **constant** across individuals, random effects **vary**.
2. Effects are fixed if they are **interesting** in themselves; random if interest in the underlying population.

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3. Fixed when sample **exhausts** the population; random when the sample is small part of the population.

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Random vs Fixed effects?

1. Fixed effects **constant** across individuals, random effects **vary**.
2. Effects are fixed if they are **interesting** in themselves; random if interest in the underlying population.
3. Fixed when sample **exhausts** the population; random when the sample is small part of the population.
4. Random effect if it's assumed to be a realized value of **random variable**.
5. Fixed effects estimated using least squares or maximum likelihood; random effects estimated with **shrinkage**.

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What is a random effect, really?

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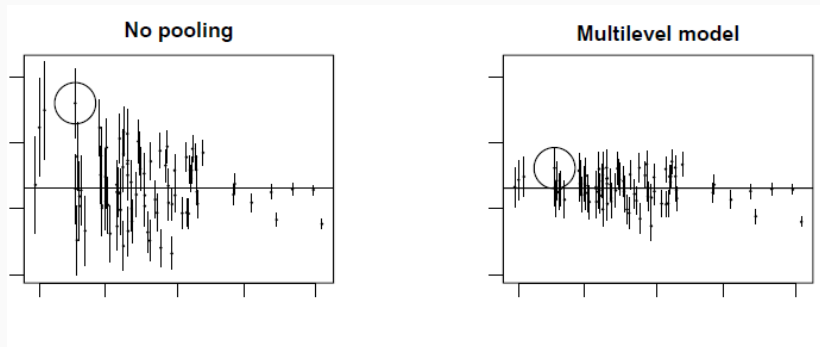
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- Variation estimated with **probability model**

Random effects are estimated with *partial pooling* (shrinkage, regularisation), while fixed effects are not (infinite variance).

Shrinkage improves parameter estimation

Especially for groups with low sample size



From Gelman & Hill p. 253

Fitting mixed/multilevel models

```
library('glmmTMB')  
mixed <- glmmTMB(height ~ dbh + (1|site), data = trees)
```

```
Family: gaussian ( identity )  
Formula:      height ~ dbh + (1 | site)  
Data: trees
```

	AIC	BIC	logLik	-2*log(L)	df.resid
	5110.3	5129.9	-2551.1	5102.3	996

Random effects:

Conditional model:

Groups	Name	Variance	Std.Dev.
site	(Intercept)	10.007	3.163
Residual		9.252	3.042

Number of obs: 1000, groups: site, 10

Dispersion estimate for gaussian family (sigma^2): 9.25

Conditional model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	19.014671	1.045018	18.20	<2e-16 ***
dbh	0.616911	0.007569	81.51	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Fitting mixed/multilevel models

```
library('lme4')
mixed <- lmer(height ~ dbh + (1|site), data = trees)
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: height ~ dbh + (1 | site)
Data: trees
```

```
REML criterion at convergence: 5108.3
```

```
Scaled residuals:
```

	Min	1Q	Median	3Q	Max
	-3.3199	-0.6607	0.0227	0.6716	3.7328

```
Random effects:
```

Groups	Name	Variance	Std.Dev.
site	(Intercept)	11.195	3.346
Residual		9.261	3.043

```
Number of obs: 1000, groups: site, 10
```

```
Fixed effects:
```

	Estimate	Std. Error	t value
(Intercept)	19.011468	1.100444	17.28
dbh	0.616927	0.007572	81.47

```
Correlation of Fixed Effects:
```

```
(Intr)
dbh -0.197
```

```
library(equatiomatic)  
equatiomatic::extract_eq(mixed)
```

$$\begin{aligned} \text{height}_i &\sim N(\alpha_{j[i]} + \beta_1(\text{dbh}), \sigma^2) \\ \alpha_j &\sim N(\mu_{\alpha_j}, \sigma_{\alpha_j}^2), \text{ for site } j = 1, \dots, J \end{aligned} \tag{1}$$


```
coef(mixed)
```

```
$site
```

```
  (Intercept)      dbh
```

1	16.70800	0.6169271
2	23.19162	0.6169271
3	21.04229	0.6169271
4	18.64086	0.6169271
5	20.32995	0.6169271
6	20.88200	0.6169271
7	16.61686	0.6169271
8	11.88302	0.6169271
9	21.84779	0.6169271
10	18.97228	0.6169271

```
attr(,"class")
```

```
[1] "coef.mer"
```

Compare site effects between mixed and lm

	lm	mixed
site1	16.7	16.7
site2	23.2	23.2
site3	21.1	21.0
site4	18.6	18.6
site5	20.3	20.3
site6	20.9	20.9
site7	16.5	16.6
site8	11.4	11.9
site9	22.1	21.8
site10	19.0	19.0

Broom: model estimates in tidy form

```
library(broom.mixed)
tidy(mixed)
```

```
# A tibble: 4 x 6
```

	effect	group	term	estimate	std.error	statistic
	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>
1	fixed	<NA>	(Intercept)	19.0	1.10	17.3
2	fixed	<NA>	dbh	0.617	0.00757	81.5
3	ran_pars	site	sd__(Intercept)	3.35	NA	NA
4	ran_pars	Residual	sd__Observation	3.04	NA	NA

See [broom.mixed](#)

Results table

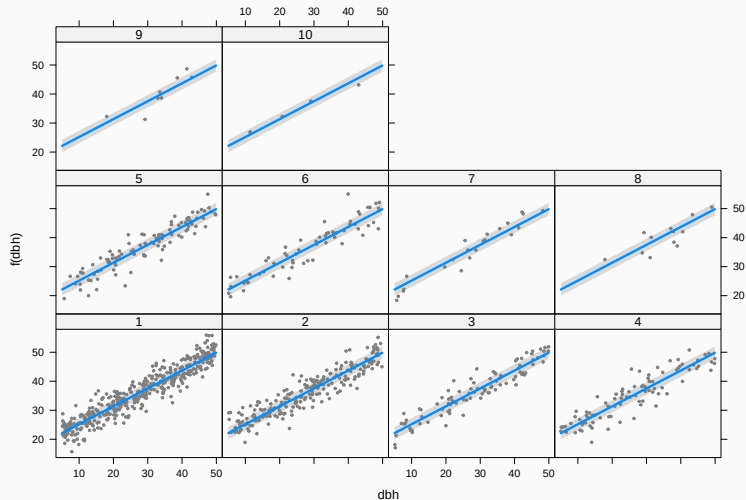
```
library(modelsummary)
```

```
modelsummary(mixed)
```

	(1)
(Intercept)	19.011 (1.100)
dbh	0.617 (0.008)
SD (Intercept site)	3.346
SD (Observations)	3.043
Num.Obs.	1000
R2 Marg.	0.753
R2 Cond.	0.888
AIC	5116.3
BIC	5135.9
ICC	0.5
RMSE	3.03

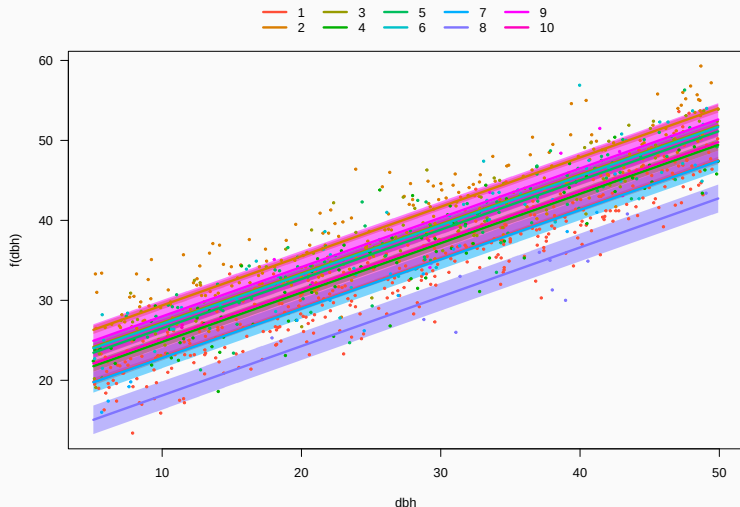
Visualising model: visreg

```
visreg(mixed, xvar = 'dbh', by = 'site')
```



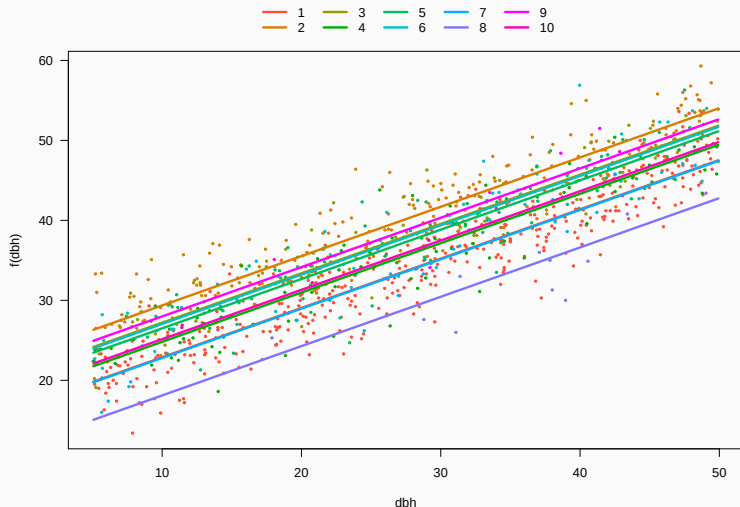
Visualising model

```
visreg(mixed, xvar = 'dbh', by = 'site', overlay = TRUE)
```



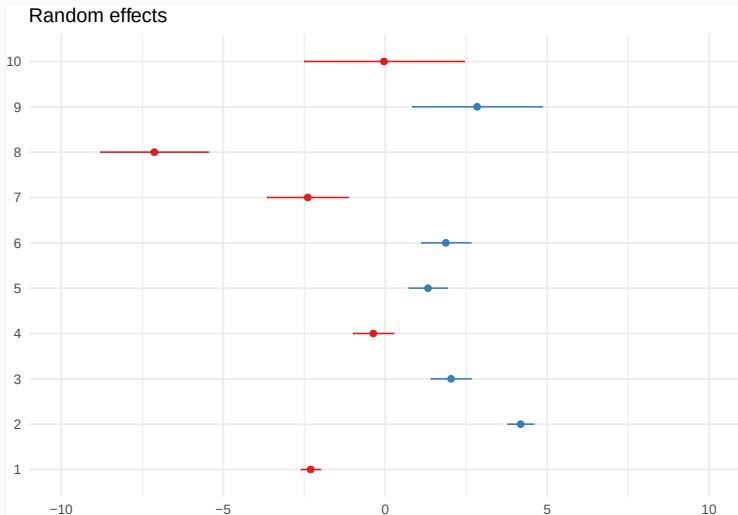
Visualising model

```
visreg(mixed, xvar = 'dbh', by = 'site', overlay = TRUE, band = FALSE)
```



Visualising model: sjPlot

```
sjPlot::plot_model(mixed, type = 're')
```

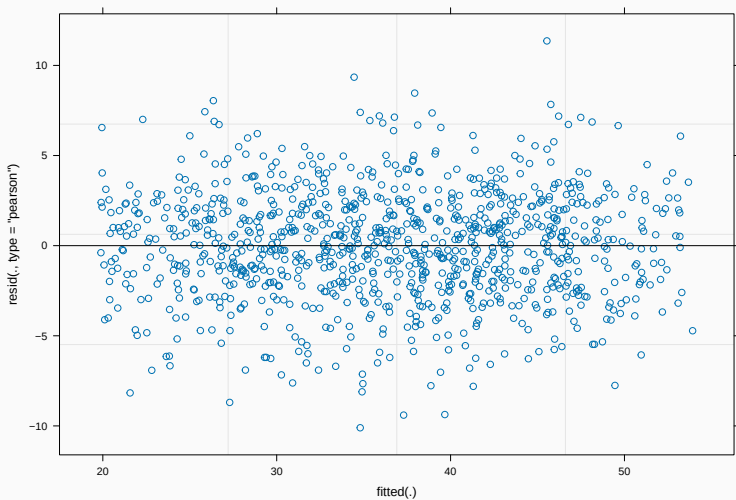


Using merTools to understand fitted model

```
library('merTools')  
shinyMer(mixed)
```

Checking residuals

```
plot(mixed)
```

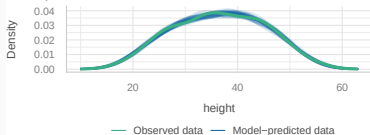


Checking residuals

```
library('performance')  
check_model(mixed)
```

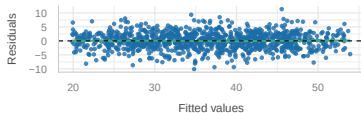
Posterior Predictive Check

Model-predicted lines should resemble observed data line



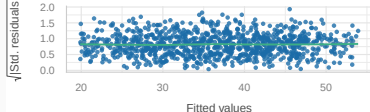
Linearity

Reference line should be flat and horizontal



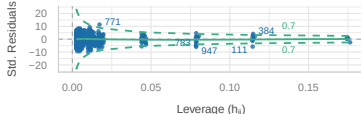
Homogeneity of Variance

Reference line should be flat and horizontal



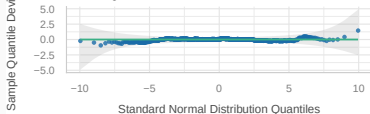
Influential Observations

Points should be inside the contour lines



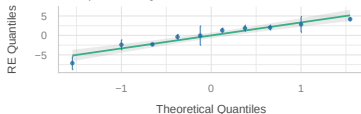
Normality of Residuals

Points should fall along the line



Normality of Random Effects (site)

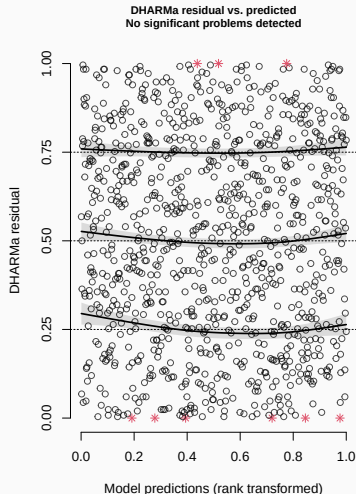
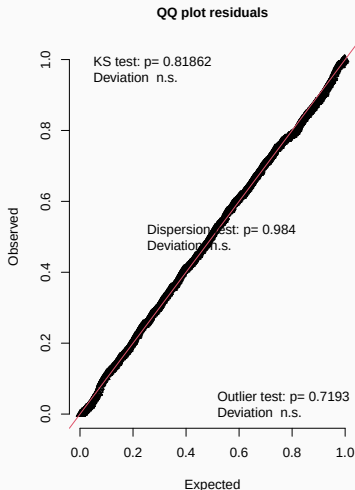
Dots should be plotted along the line



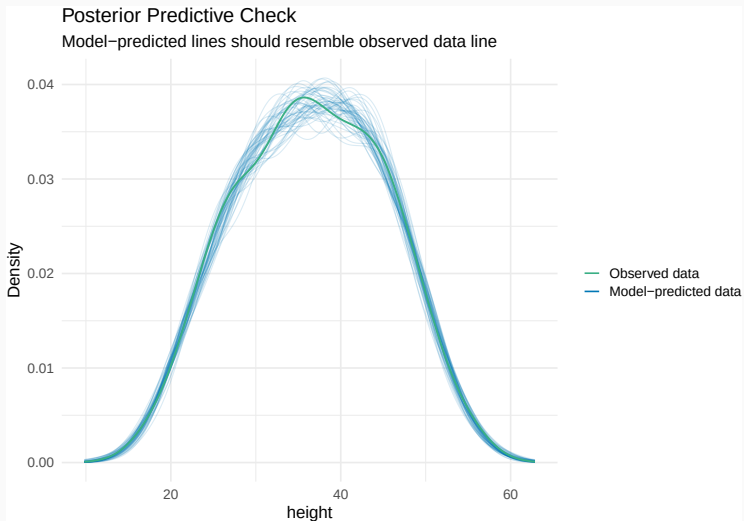
Checking residuals (DHARMA)

```
DHARMA::simulateResiduals(mixed, plot = TRUE, re.form = NULL)
```

DHARMA residual



```
check_predictions(mixed)
```



Many approaches! Somewhat polemic (e.g. see [this](#)).

Nakagawa & Schielzeth propose **marginal** (considering fixed effects only) and **conditional** R^2 (including random effects too):

```
r2(mixed)
```

```
# R2 for Mixed Models
```

```
Conditional R2: 0.888
```

```
  Marginal R2: 0.753
```

Growing the hierarchy: adding site-level predictors

We had:

$$y_i = a + \alpha_j + b \cdot x_i + \varepsilon_i$$

$$\alpha_j \sim N(0, \tau^2)$$

$$\varepsilon_i \sim N(0, \sigma^2)$$

We had:

$$y_i = a + \alpha_j + b \cdot x_i + \varepsilon_i$$

$$\alpha_j \sim N(0, \tau^2)$$

$$\varepsilon_i \sim N(0, \sigma^2)$$

Now

$$y_i = a + \alpha_j + b \cdot x_i + \varepsilon_i$$

$$\alpha_j \sim N(\mu_j, \tau^2)$$

$$\mu_j = \delta \cdot \text{Predictor}_j$$

$$\varepsilon_i \sim N(0, \sigma^2)$$

Are height differences among sites related to temperature?

$$\text{Height}_i = \text{site}_j + b \cdot \text{DBH}_i + \varepsilon_i$$

$$\text{site}_j \sim N(\mu_j, \tau^2)$$

$$\mu_j = a + \delta \cdot \text{Temperature}_j$$

$$\varepsilon_i \sim N(0, \sigma^2)$$

Are height differences among sites related to temperature?

```
sitedata <- read.csv('data/sitedata.csv')  
sitedata
```

	site	temp
1	1	15.1
2	2	22.0
3	3	20.1
4	4	20.4
5	5	20.0
6	6	20.1
7	7	17.5
8	8	14.6
9	9	19.2
10	10	16.0

```
trees.full <- merge(trees, sitedata, by = 'site')  
head(trees.full)
```

	site	dbh	height	sex	dead	temp
1	1	21.05	32.2	male	0	15.1
2	1	46.63	45.9	female	0	15.1
3	1	43.86	45.5	male	0	15.1
4	1	29.03	35.5	male	0	15.1
5	1	6.02	21.1	male	0	15.1
6	1	40.82	38.7	male	0	15.1

Fit multilevel model

```
group.pred <- lmer(height ~ dbh + (1 | site) + temp, data = trees.full)
```

Linear mixed model fit by REML ['lmerMod']

Formula: height ~ dbh + (1 | site) + temp

Data: trees.full

REML criterion at convergence: 5098.2

Scaled residuals:

	Min	1Q	Median	3Q	Max
	-3.3247	-0.6517	0.0192	0.6663	3.7268

Random effects:

Groups	Name	Variance	Std.Dev.
site	(Intercept)	3.158	1.777
Residual		9.266	3.044

Number of obs: 1000, groups: site, 10

Fixed effects:

	Estimate	Std. Error	t value
(Intercept)	-1.730910	4.671330	-0.371
dbh	0.616894	0.007571	81.484
temp	1.115104	0.248000	4.496

Correlation of Fixed Effects:

	(Intr) dbh
dbh	-0.055
temp	-0.991 0.008

Too strong correlation of parameters!

Centre (and scale) continuous variables

```
mean(sitedata$temp)
```

```
[1] 18.5
```

```
trees.full$temp.c <- trees.full$temp - 18
```

Temperatures now referred as deviations from 18 °C (close to average)

Fit multilevel model

```
group.pred <- lmer(height ~ dbh + (1 | site) + temp.c, data = trees.full)
```

Linear mixed model fit by REML ['lmerMod']

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Data: trees.full

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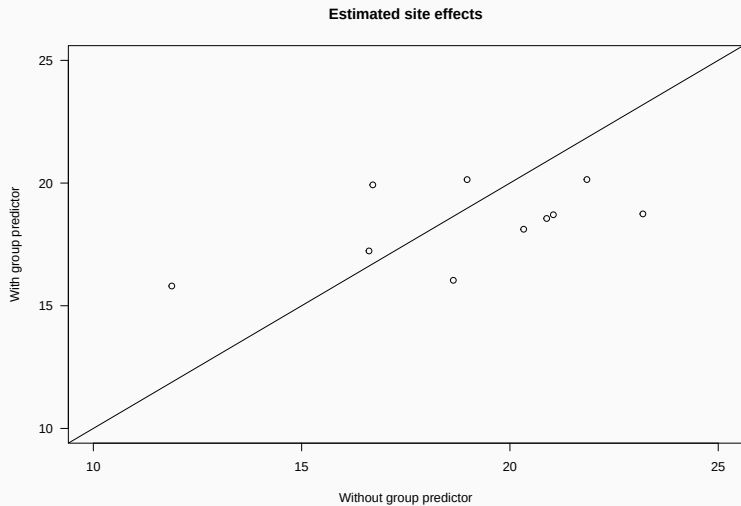
	Estimate	Std. Error	t value
(Intercept)	18.340954	0.655054	27.999
dbh	0.616894	0.007571	81.484
temp.c	1.115104	0.248000	4.496

Correlation of Fixed Effects:

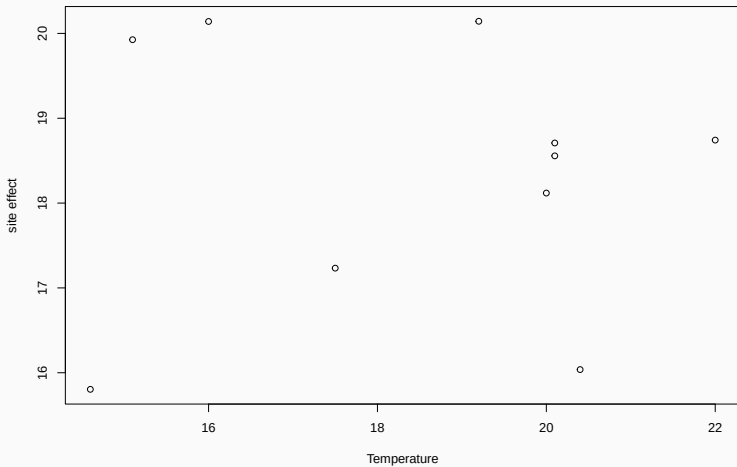
(Intr) dbh

```
shinyMer(group.pred)
```


Comparing site effects with and without group predictor



Are site effects related to temperature?



Results table

```
modelsummary(group.pred)
```

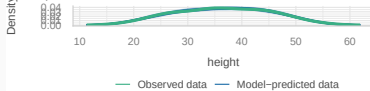
	(1)
(Intercept)	18.341
	(0.655)
dbh	0.617
	(0.008)
temp.c	1.115
	(0.248)
SD (Intercept site)	1.777
SD (Observations)	3.044
Num.Obs.	1000
R2 Marg.	0.853
R2 Cond.	0.890
AIC	5108.2
BIC	5132.7

Model checking

```
check_model(group.pred)
```

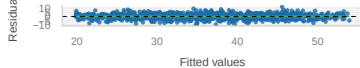
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Model-predicted lines should resemble observed data line



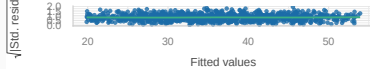
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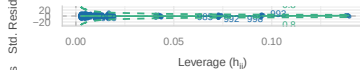
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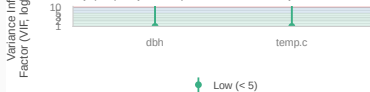
Influential Observations

Points should be inside the contour lines



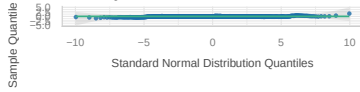
Collinearity

High collinearity (VIF) may inflate parameter uncertainty



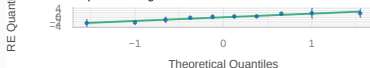
Normality of Residuals

Dots should fall along the line



Normality of Random Effects (site)

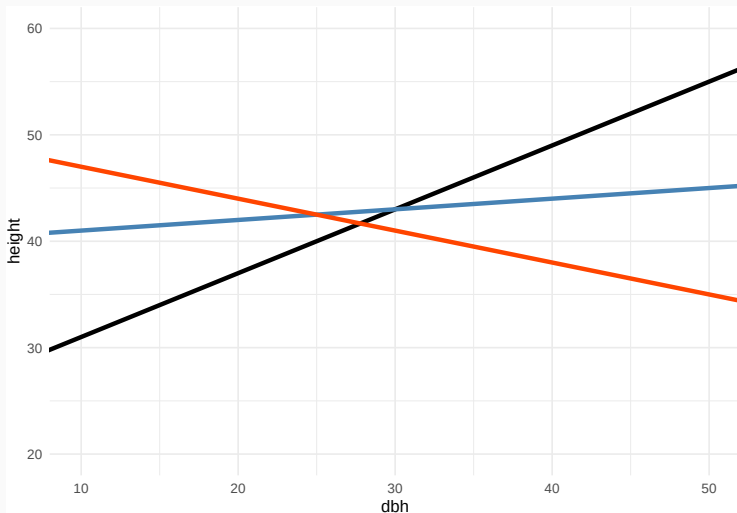
Dots should be plotted along the line



Varying intercepts and slopes

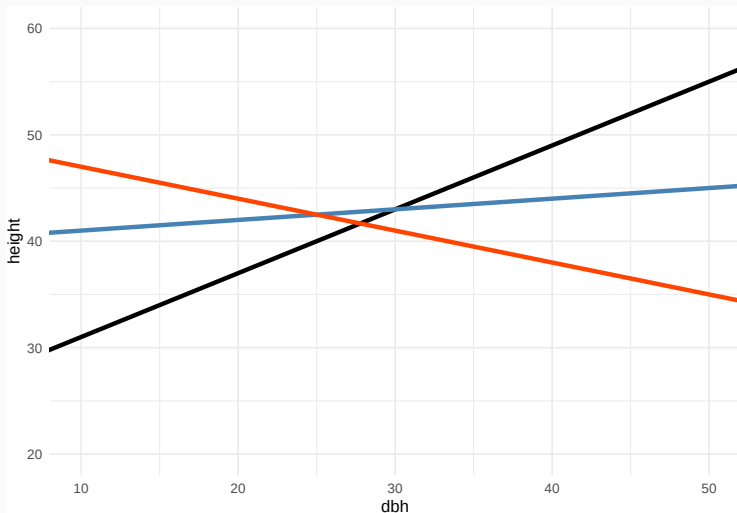
Varying intercepts and slopes

- There is overall **difference in height** among sites (different intercepts)



Varying intercepts and slopes

- There is overall **difference in height** among sites (different intercepts)
- **Relationship** between DBH and Height **varies** among sites (different slopes)



Varying intercepts and slopes

```
mixed.slopes <- lmer(height ~ dbh + (1 + dbh | site), data=trees)
equatiomatic::extract_eq(mixed.slopes)
```

$$\begin{aligned} \text{height}_i &\sim N(\alpha_{j[i]} + \beta_{1j[i]}(\text{dbh}), \sigma^2) \\ \begin{pmatrix} \alpha_j \\ \beta_{1j} \end{pmatrix} &\sim N\left(\begin{pmatrix} \mu_{\alpha_j} \\ \mu_{\beta_{1j}} \end{pmatrix}, \begin{pmatrix} \sigma_{\alpha_j}^2 & \rho_{\alpha_j\beta_{1j}} \\ \rho_{\beta_{1j}\alpha_j} & \sigma_{\beta_{1j}}^2 \end{pmatrix}\right), \text{ for site } j = 1, \dots, J \end{aligned} \quad (2)$$

Varying intercepts and slopes

Linear mixed model fit by REML ['lmerMod']

Formula: height ~ dbh + (1 + dbh | site)

Data: trees

REML criterion at convergence: 5105.1

Scaled residuals:

Min	1Q	Median	3Q	Max
-3.3342	-0.6599	0.0375	0.6916	3.7756

Random effects:

Groups	Name	Variance	Std.Dev.	Corr
site	(Intercept)	1.566e+01	3.95671	
	dbh	3.087e-04	0.01757	-1.00
Residual		9.226e+00	3.03744	

Number of obs: 1000, groups: site, 10

Fixed effects:

	Estimate	Std. Error	t value
(Intercept)	18.95272	1.29190	14.67
dbh	0.61837	0.00946	65.37

Correlation of Fixed Effects:

(Intr)

dbh -0.722

optimizer (nloptwrap) convergence code: 0 (OK)

Varying intercepts and slopes

```
$site
      (Intercept)      dbh
1      16.34655 0.6299443
2      23.74733 0.5970814
3      21.28802 0.6080019
4      18.57844 0.6200337
5      20.47961 0.6115916
6      21.09608 0.6088542
7      16.17675 0.6306983
8      10.54681 0.6556978
9      22.27301 0.6036281
10     18.99463 0.6181856
```

```
attr(,"class")
[1] "coef.mer"
```

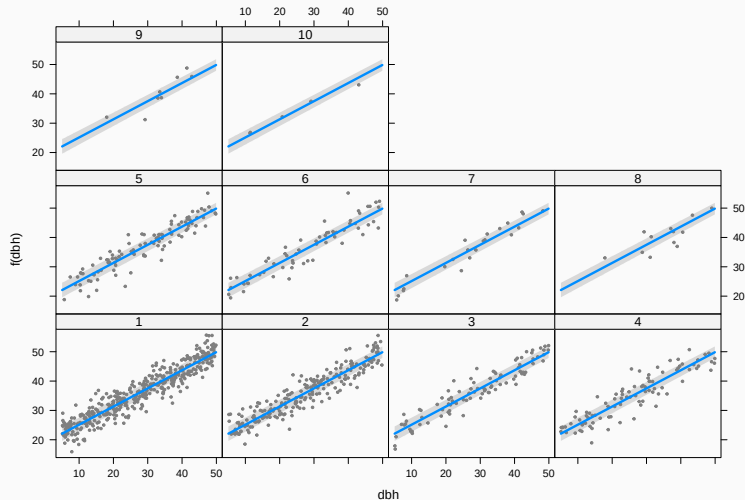
Results table

```
modelsummary(mixed.slopes)
```

	(1)
(Intercept)	18.953
	(1.292)
dbh	0.618
	(0.009)
SD (Intercept site)	3.957
SD (dbh site)	0.018
Cor (Intercept dbh site)	-
	1.000
SD (Observations)	3.037
Num.Obs.	1000
R2 Marg.	0.871
AIC	5117.1
BIC	5146.5
RMSE	3.02

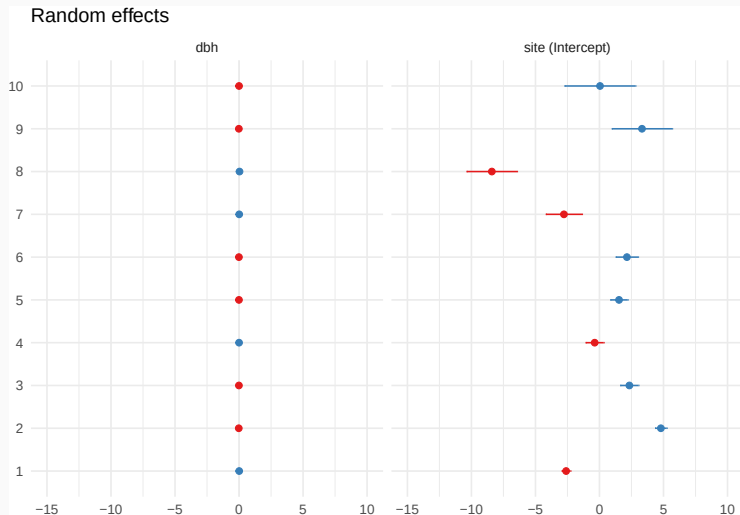
Visualising model: visreg

```
visreg(mixed.slopes, xvar = "dbh", by = "site")
```



Visualising model: sjPlot

```
plot_model(mixed.slopes, type = 're')
```

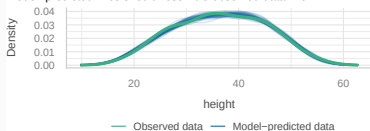


Model checking

```
check_model(mixed.slopes)
```

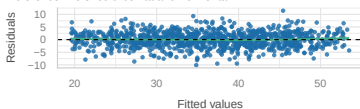
Posterior Predictive Check

Model-predicted lines should resemble observed data line



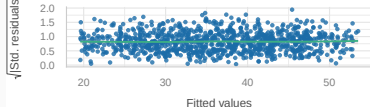
Linearity

Reference line should be flat and horizontal



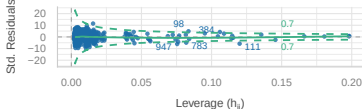
Homogeneity of Variance

Reference line should be flat and horizontal



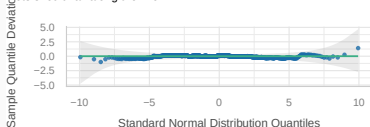
Influential Observations

Points should be inside the contour lines



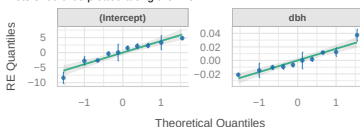
Normality of Residuals

Dots should fall along the line



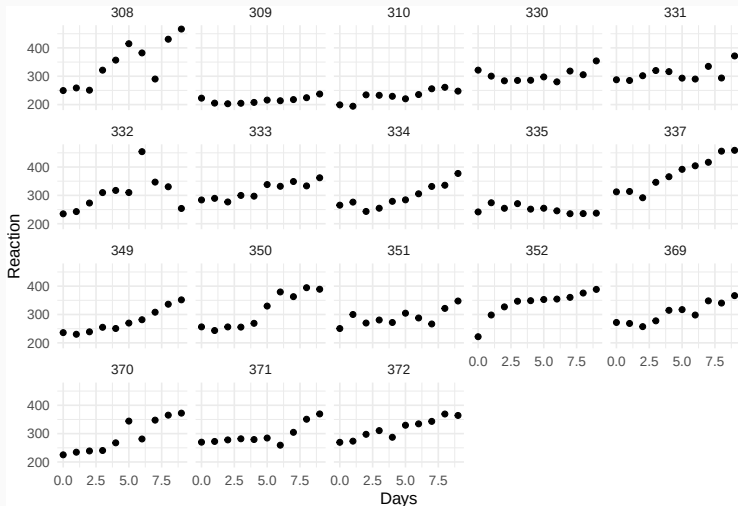
Normality of Random Effects (site)

Dots should be plotted along the line



More examples

sleepstudy (repeated measures)



Varying intercepts and slopes (lme4)

```
sleep <- lmer(Reaction ~ Days + (1+Days|Subject), data = sleepstudy)
```

Linear mixed model fit by REML ['lmerMod']

Formula: Reaction ~ Days + (1 + Days | Subject)

Data: sleepstudy

REML criterion at convergence: 1743.6

Scaled residuals:

Min	1Q	Median	3Q	Max
-3.9536	-0.4634	0.0231	0.4634	5.1793

Random effects:

Groups	Name	Variance	Std.Dev.	Corr
Subject	(Intercept)	612.10	24.741	
	Days	35.07	5.922	0.07
Residual		654.94	25.592	

Number of obs: 180, groups: Subject, 18

Fixed effects:

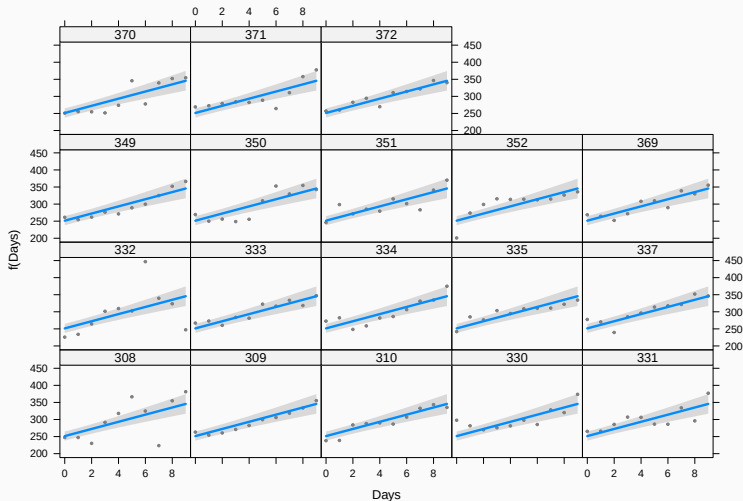
	Estimate	Std. Error	t value
(Intercept)	251.405	6.825	36.838
Days	10.467	1.546	6.771

Correlation of Fixed Effects:

(Intr)

Varying intercepts and slopes (lme4)

```
visreg(sleep, xvar = 'Days', by = 'Subject')
```



Fitting multilevel models (GAMM) with mgcv

```
sgamm <- mgcv::gam(Reaction ~ s(Days, Subject, k = 3, bs = 'fs'),  
  data = sleepstudy, method = 'REML')
```

Family: gaussian

Link function: identity

Formula:

Reaction ~ s(Days, Subject, k = 3, bs = "fs")

Parametric coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	298.51	9.05	32.98	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:

	edf	Ref.df	F	p-value
s(Days,Subject)	45.67	53	17.11	<2e-16 ***

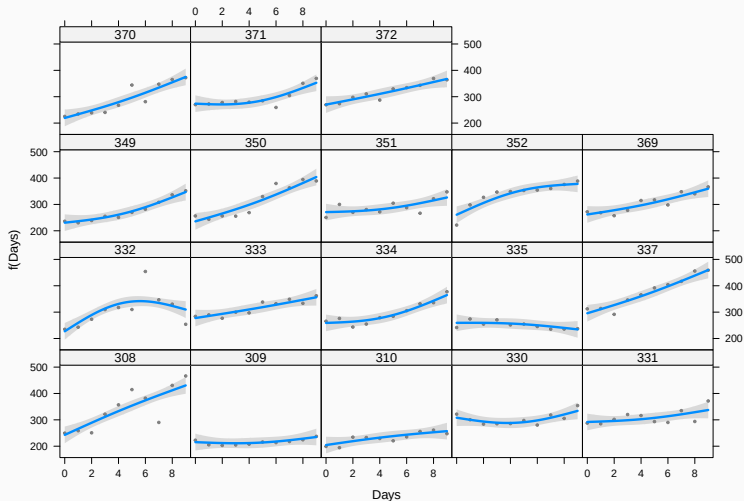
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) = 0.835 Deviance explained = 87.7%

-REML = 883.64 Scale est. = 523.2 n = 180

Fitting multilevel models (GAMM) with mgcv

```
visreg(sgam, xvar = 'Days', by = 'Subject')
```



Hierarchical generalized additive models: an introduction with mgcv

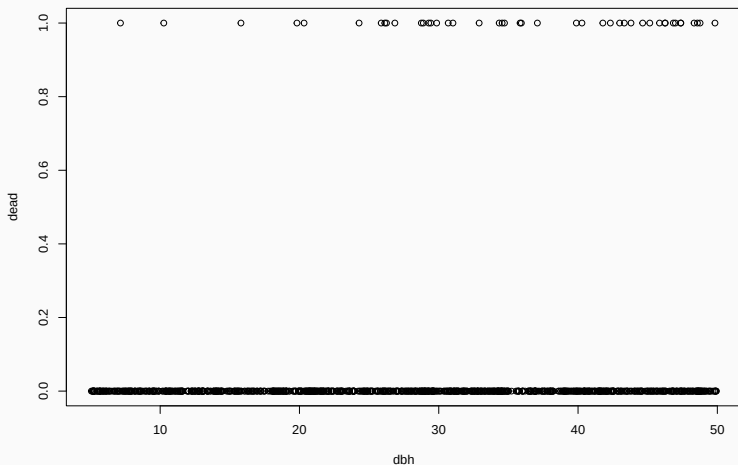
Eric J Pedersen Corresp., 1, 2, David L. Miller 3, 4, Gavin L. Simpson 5, Noam Ross 6

<https://doi.org/10.7287/peerj.preprints.27320v1>

Multilevel logistic regression

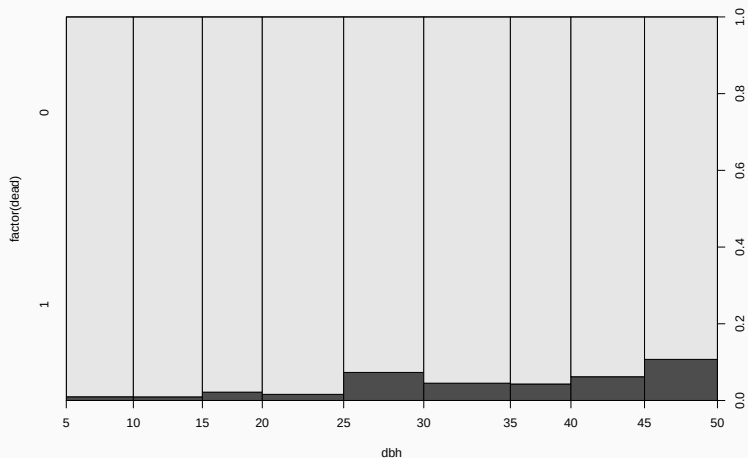
Q: Relationship between tree size and mortality

```
plot(dead ~ dbh, data = trees)
```



Q: Relationship between tree size and mortality

```
plot(factor(dead) ~ dbh, data = trees)
```



Fit simple logistic regression

```
simple.logis <- glm(dead ~ dbh, data = trees, family=binomial)
```

Call:

```
glm(formula = dead ~ dbh, family = binomial, data = trees)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-4.77874	0.50902	-9.388	< 2e-16 ***
dbh	0.05365	0.01377	3.895	9.82e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 360.91 on 999 degrees of freedom
Residual deviance: 343.69 on 998 degrees of freedom
AIC: 347.69

Number of Fisher Scoring iterations: 6

Logistic regression with *independent* site effects

```
logis2 <- glm(dead ~ dbh + site, data = trees, family=binomial)
```

Call:

```
glm(formula = dead ~ dbh + site, family = binomial, data = trees)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-4.80123	0.54985	-8.732	<2e-16 ***
dbh	0.05371	0.01381	3.889	0.0001 ***
site2	-0.29692	0.46073	-0.644	0.5193
site3	0.21275	0.52799	0.403	0.6870
site4	0.39841	0.53025	0.751	0.4524
site5	-0.42557	0.64018	-0.665	0.5062
site6	0.66861	0.53656	1.246	0.2127
site7	0.11862	1.06211	0.112	0.9111
site8	0.43899	1.08058	0.406	0.6846
site9	-13.63389	840.90382	-0.016	0.9871
site10	-13.17148	1042.21823	-0.013	0.9899

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 360.91 on 999 degrees of freedom
Residual deviance: 338.58 on 989 degrees of freedom

Fit multilevel logistic regression

```
mixed.logis <- glmer(dead ~ dbh + (1|site), data=trees, family = binomial)
```

Generalized linear mixed model fit by maximum likelihood (Laplace

Approximation) [glmerMod]

Family: binomial (logit)

Formula: dead ~ dbh + (1 | site)

Data: trees

AIC	BIC	logLik	-2*log(L)	df.resid
349.7	364.4	-171.8	343.7	997

Scaled residuals:

Min	1Q	Median	3Q	Max
-0.3498	-0.2528	-0.1888	-0.1370	9.0031

Random effects:

Groups	Name	Variance	Std.Dev.
site	(Intercept)	0	0

Number of obs: 1000, groups: site, 10

Fixed effects:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-4.77874	0.50904	-9.388	< 2e-16 ***
dbh	0.05365	0.01377	3.895	9.83e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

(Intr)

dbh -0.952

optimizer (Nelder_Mead) convergence code: 0 (OK)

boundary (singular) fit: see help('isSingular')

Retrieve model coefficients

```
coef(mixed.logis)
```

```
$site
```

```
  (Intercept)      dbh
```

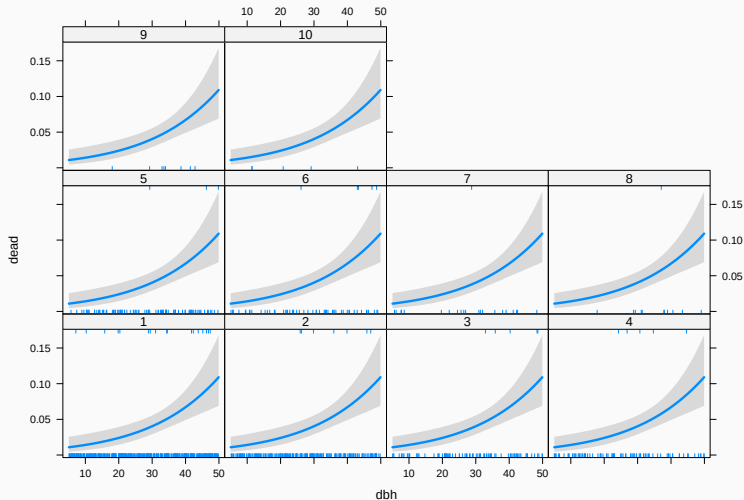
1	-4.778744	0.05364989
2	-4.778744	0.05364989
3	-4.778744	0.05364989
4	-4.778744	0.05364989
5	-4.778744	0.05364989
6	-4.778744	0.05364989
7	-4.778744	0.05364989
8	-4.778744	0.05364989
9	-4.778744	0.05364989
10	-4.778744	0.05364989

```
attr(,"class")
```

```
[1] "coef.mer"
```

Visualising model: visreg

```
visreg(mixed.logis, xvar = 'dbh', by = 'site', scale = 'response')
```



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- Predictors enter at the appropriate level

Advantages of multilevel models

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- More **efficient inference** of regression parameters
- Using all the data to perform inferences for groups with **small sample size**
- Allow predictions for **unobserved groups**

- Varying intercepts

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 - $y \sim x + (1 \mid \text{group})$

- Varying intercepts
 - $y \sim x + (1 \mid \text{group})$
- Varying intercepts and slopes

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- Varying intercepts, 2 groups (crossed)

Formula syntax for different models

- Varying intercepts
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- Varying intercepts and slopes
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- Varying intercepts, 2 groups (crossed)
 - $y \sim x + (1 \mid \text{group1}) + (1 \mid \text{group2})$

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 - $y \sim x + (1 \mid \text{group/subgroup})$
 - This is **equivalent** to $y \sim x + (1 \mid \text{group1}) + (1 \mid \text{group2})$ with distinct labelling of group levels.

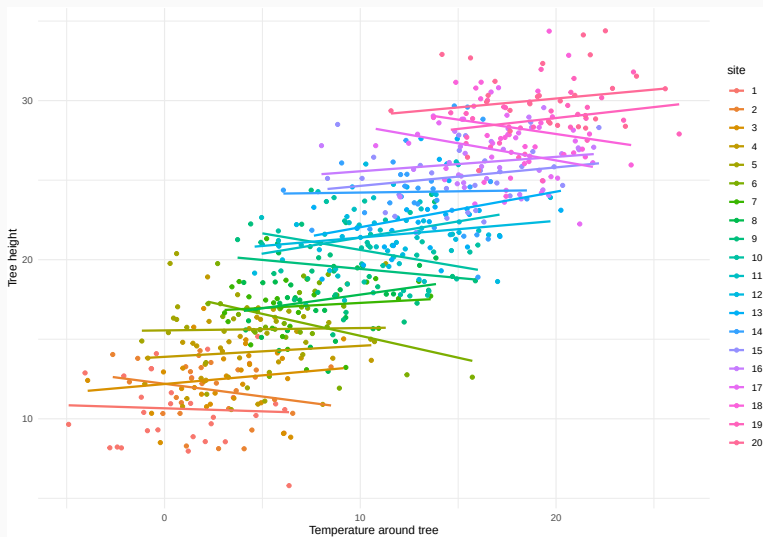
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Formula syntax for different models

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- Varying intercepts and slopes, 2 groups (crossed)
 - $y \sim x + (1 + x \mid \text{group1}) + (1 + x \mid \text{group2})$

Heterogeneity bias

Predictor (temperature) correlates with groups (random site effects)



Fitting a model with varying intercepts

```
Family: gaussian ( identity )  
Formula:          tree.height ~ tree.temp + (1 | site)  
Data: df
```

AIC	BIC	logLik	-2*log(L)	df.resid
2681.6	2699.2	-1336.8	2673.6	596

Random effects:

Conditional model:

Groups	Name	Variance	Std.Dev.
site	(Intercept)	31.119	5.578
Residual		4.212	2.052

Number of obs: 600, groups: site, 20

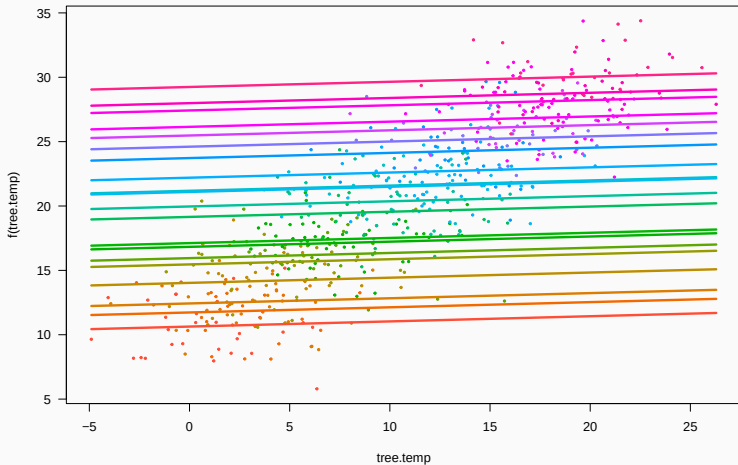
Dispersion estimate for gaussian family (σ^2): 4.21

Conditional model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	20.11896	1.28855	15.614	<2e-16 ***
tree.temp	0.04021	0.02954	1.361	0.173

Fitting a model with varying intercepts

```
visreg(m, xvar = "tree.temp", by = "site", overlay = T, band = F, legend = F)
```



Fitting a model with varying intercepts and slopes

```
Family: gaussian ( identity )  
Formula:          tree.height ~ tree.temp + (1 + tree.temp | site)  
Data: df
```

AIC	BIC	logLik	-2*log(L)	df.resid
2684.7	2711.1	-1336.4	2672.7	594

Random effects:

Conditional model:

Groups	Name	Variance	Std.Dev.	Corr
site	(Intercept)	2.813e+01	5.30422	
	tree.temp	7.845e-04	0.02801	0.99
Residual		4.205e+00	2.05054	

Number of obs: 600, groups: site, 20

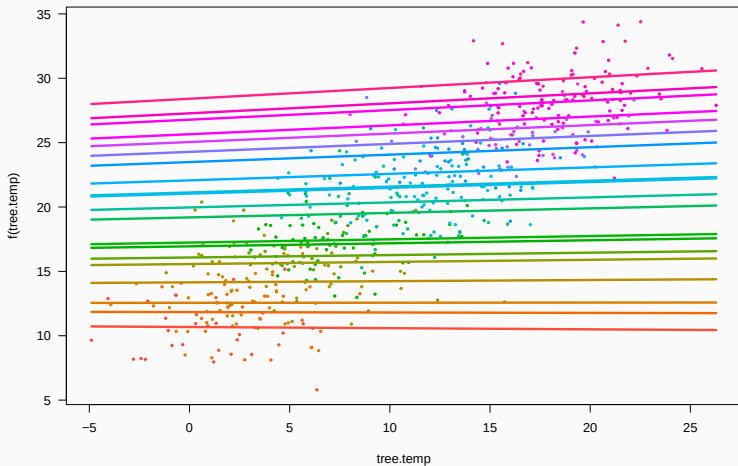
Dispersion estimate for gaussian family (σ^2): 4.2

Conditional model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	19.96903	1.24001	16.104	<2e-16 ***
tree.temp	0.03936	0.03021	1.303	0.193

Fitting a model with varying intercepts and slopes

```
visreg(m, xvar = "tree.temp", by = "site", overlay = T, band = F, legend = F)
```



Do predictors vary within- and between groups?

```
performance::check_group_variation(m)
```

Check site variation

Variable	Variation	Design
----------	-----------	--------

-------	--	--

tree.temp	both	
-----------	------	--

Group-mean centering

```
df <- datawizard::demean(df, select = "tree.temp", by = "site")  
head(df)
```

	site	tree	tree.temp	tree.height	tree.temp_between	tree.temp_within
1	1	1	-0.6814269	13.148025	0.8586887	-1.54011
2	1	2	0.3094675	10.945306	0.8586887	-0.54922
3	1	3	5.6761249	10.933339	0.8586887	4.81743
4	1	4	1.2115252	7.967865	0.8586887	0.35283
5	1	5	1.3878632	12.580771	0.8586887	0.52917
6	1	6	6.1451950	10.578532	0.8586887	5.28650

Model with site effects and temperature (between & within)

```
m <- glmmTMB(tree.height ~ tree.temp_between + tree.temp_within + (1|site), data = df)
```

Family: gaussian (identity)

Formula:

tree.height ~ tree.temp_between + tree.temp_within + (1 | site)

Data: df

AIC	BIC	logLik	-2*log(L)	df.resid
2589.1	2611.1	-1289.6	2579.1	595

Random effects:

Conditional model:

Groups	Name	Variance	Std.Dev.
site	(Intercept)	0.1429	0.3781
Residual		4.2093	2.0516

Number of obs: 600, groups: site, 20

Dispersion estimate for gaussian family (sigma^2): 4.21

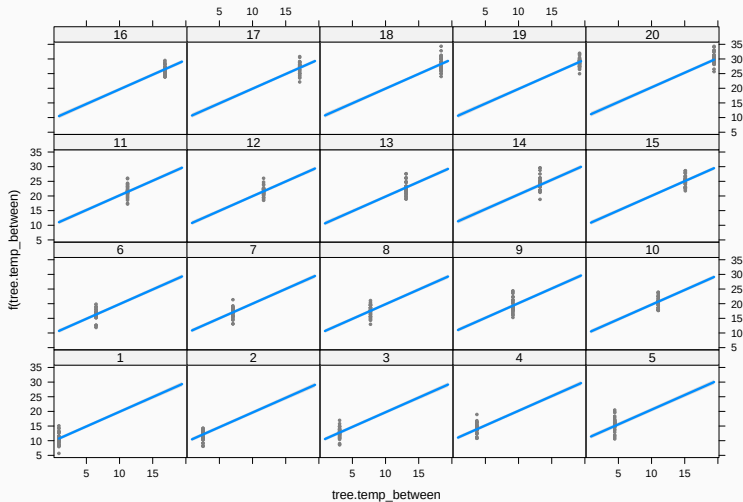
Conditional model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	9.96527	0.24758	40.25	<2e-16 ***
tree.temp_between	1.00125	0.02055	48.73	<2e-16 ***
tree.temp_within	0.02256	0.02925	0.77	0.44

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

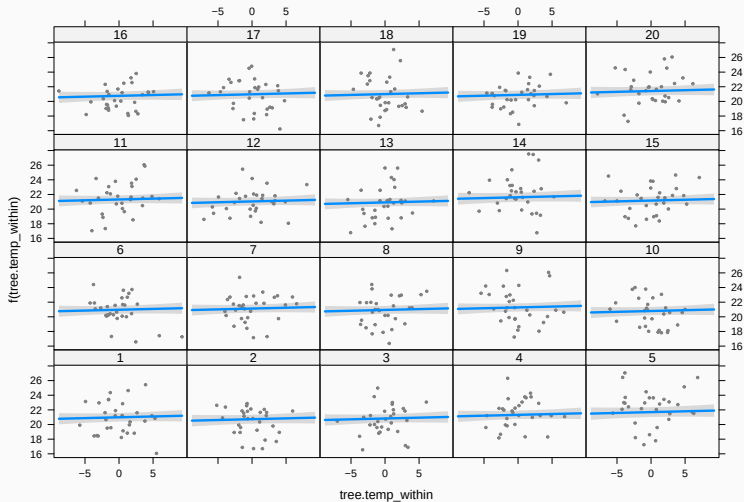
Model with site effects and temperature (between & within)

```
visreg(m, xvar = "tree.temp_between", by = "site")
```



Model with site effects and temperature (between & within)

```
visreg(m, xvar = "tree.temp_within", by = "site")
```



<https://bbolker.github.io/mixedmodels-misc/glmmFAQ.html>

<http://mfviz.com/hierarchical-models/>

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